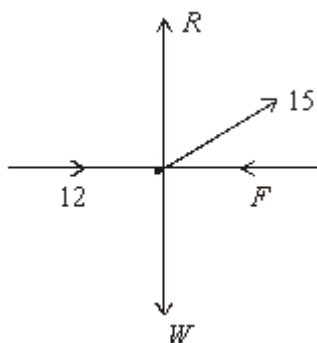


Mark schemes

Q1.

(a)



any 3 correct, including arrows and labels

B1

all correct, including arrows and labels

B1

2

accept mg for W

*accept either 15 N or components, but **not** both*

(b) $F = 12 + 15 \cos 30^\circ$

for M1, both terms on RHS and sin/cos

M1A1

$= 12 + 12.99$

$F = 25 \text{ N}$

CAO

A1

3

(c) $25 = \frac{1}{3} R$

M1A0 if inequality used

M1

$R = 75 \text{ N}$

AWRT; ft F; accept use of c's F from (b);

accept 0.333 for μ

A1F

2

(b) $R + 15\cos 60^\circ = W$
3 terms and attempt at component
 $Wg \Rightarrow M1A0A0$

M1

$$75 + 7 \frac{1}{2} = W$$

ft R

A1F

$$W = 82 \frac{1}{2} \text{ N}$$

AWRT 82.5, ft R or one slip

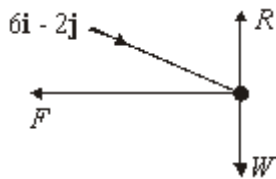
A1F

3

[10]

Q2.

(a)



B1

1

(b) $6 - F = 1.5 \times 0.6$
3 terms for M1

M1 A1

$$F = 5.1\text{N}$$

CAO

A1

3

(c) $R = 2 + 1.5 \times 9.8$
RHS, 2 forces for M1

M1A1F

$$R = 16.7\text{N}$$

A1F

3

(d) $F = \mu R$
 $5.1 = \mu \times 16.7$

M1

$\mu = 0.305$

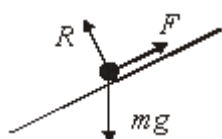
A1F

2

[9]

Q3.

(a) (i)



Correct force diagram

B1

1

(ii) $R = 20 \times 9.8 \cos 40^\circ = 150 \text{ N}$
Resolving perpendicular to the slope

M1

ag Correct answer from correct working

A1

2

(iii) $F = 20 \times 9.8 \sin 40^\circ = 126 \text{ N}$
Resolving parallel to the slope

M1

$126 = \mu \times 150$
Correct friction

A1

$\mu = \frac{126}{150} = 0.84$
Use of $F = \mu R$

M1

ag Correct μ from correct working

A1

4

- (b) It would be reduced
Correct explanation

B1

1

[8]

Q4.

(a) $R = 2 \times 9.8 \cos 40^\circ$

Resolving perpendicular to the slope

M1

$$F = 0.3 \times 2 \times 9.8 \cos 40^\circ$$

Use of $F = \mu R$

M1

$$= 4.50 \text{ N}$$

Correct F

A1

3

(b) $2a = -2 \times 9.8 \sin 40^\circ - 4.50$

Three term equation of motion

M1

$$a = \frac{-2 \times 9.8 \sin 40^\circ - 4.50}{2} = -8.55 \text{ ms}^{-2}$$

Correct acceleration

A1

3

[6]

Q5.

(a) $R = 20 \times 9.8 = 196 \text{ N}$

Finding R and using $F = \mu R$

M1

$$F = 0.15 \times 196 = 29.4 \text{ N}$$

Correct friction

A1

2

(b) $T - 29.4 = 20a$

Equation of motion for A

M1

Correct equation

A1

$5 \times 9.8 - T = 5a$

Equation of motion for B

M1

Correct equation

A1

$49 - (20a + 29.4) = 5a$

$a = \frac{19.6}{25} = 0.784 \text{ ms}^{-2}$

ag *Correct acceleration from correct working*

A1

5

(c) $T = 20 \times 0.784 + 29.4 = 45.1 \text{ N}$

Substituting a into one equation of motion

M1

Correct tension

A1

2

(d) $s = \frac{1}{2} \times 0.784 \times 3^2$

Use of constant acceleration equation with $u = 0$

M1

$= 3.53 \text{ m}$

Correct equation

A1

Correct distance

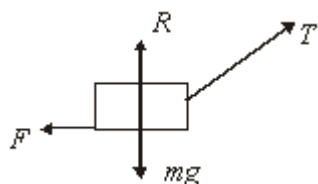
A1

3

[12]

Q6.

(a)



Correct force diagram

B1

1

(b) $R + T \sin 30^\circ = 200 \times 9.8$

Three term equation from resolving vertically

M1

$$R + 0.5 T = 1960$$

$$R = 1960 - 0.5T$$

Correct equation

A1

ag *Correct result from correct working*

A1

3

(c) $F = T \cos 30^\circ$

Resolving horizontally

M1

$$T \cos 30^\circ = 0.6(1960 - 0.5T)$$

Use of $F = \mu R$

M1

$$T (\cos 30^\circ + 0.3) = 1176$$

Correct equation

A1

Solving for T

m1

$$T = \frac{1176}{(\cos 30^\circ + 0.3)} = 1010 \text{ N}$$

Correct T

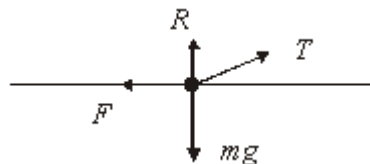
A1

5

[9]

Q7.

(a)



Correct force diagram

B1

1

(b) $R + T \sin 40^\circ = 50 \times 9.8$

$$R = 490 - T \sin 40^\circ$$

Three term equation of motion

Correct equation

M1

Correct expression for R
Use of $F R = \mu$

A1

A1

3

(c) $F = 0.6 (490 - T \sin 40^\circ)$
Use of $F = \mu R$

©2025 Exam Papers Practice. All rights reserved.

M1

$$= 294 - 0.6 T \sin 40^\circ$$

ag *Correct result from correct working*

A1

2

(d) $T \cos 40^\circ - (294 - 0.6 T \sin 40^\circ) = 50 \times 0.5$

$$T = \frac{319}{\cos 40^\circ + 0.6 \sin 40^\circ} = 277 \text{ N}$$

Four term equation of motion

M1

Correct equation

A1

Solving for T

M1

Correct T

A1

4

[10]

Q8.



B1

1

(b) $R = 5 \times 9.8 \cos 40^\circ = 37.5 \text{ N}$

Correct force diagram

Resolving perpendicular to slope

Correct R

M1 A1

2

(c) $F = 0.2R = 7.51 \text{ N}$

Using $F = \mu R$

Correct F from correct working

M1 A1

2

(d) $5 \times 9.8 \sin 40^\circ - F = 5a$

Resolving parallel to slope to give 3 term equation of motion

M1 A1

Correct equation

$$a = \frac{5 \times 9.8 \sin 40^\circ - F}{5} = 4.80 \text{ ms}^{-2}$$

m1

Solving for a

Correct a from correct working

A1

4

(e) $10^2 = 2^2 + 2 \times 4.80s$

Forming constant acceleration equation

Correct equation

M1 A1

$$s = \frac{100 - 4}{9.6} = 10.0 \text{ m}$$

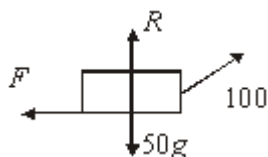
Correct s

A1

3

[12]

Q9.



(a)

Correct force diagram

B1

1

(b) $50 \times 9.8 = R + 100 \sin 30^\circ$

Resolving vertically

Correct equation

M1 A1

$R = 440 \text{ N}$

Correct F from correct working

A1

3

(c) $100 \cos 30^\circ \leq \mu \times 440$

Use of $F \leq \mu R$ or $F = \mu R$

M1

$\therefore k = 0.197$

A1

$\mu \geq 0.197$

Correct equation

Correct k from correct working

A1

(d) $50a = 100 \cos 30^\circ - 0.1 \times 440$

Resolving horizontally to obtain a 3 term equation of motion

M1

3

Correct equation

A1

Solving for a

m1

$a = 0.852 \text{ ms}^{-2}$

Correct a

Allow 0,680 or 0.681

A1

4

[11]

Q10.

(a) $R + 100 \sin 20^\circ + 60 \sin 50^\circ = 490$

Resolving vertically

Correct equation

M1 A1

$R = 410$

A1

3

(b) (i) $100 \cos 20^\circ - 60 \cos 50^\circ - F = 25$

Four term equation of motion

Correct equation

M1 A1

$F = 30.4$

Solving for F

Correct F from from correct working

M1 A1

4

(ii) $30.40 = \mu \times 409.8$

Use of $F = \mu R$

M1

$$\mu = 0.0742$$

Correct μ

A1

2

[9]

Q11.

(a) $R = 20 \times 9.8 = 196$

cao

B1

1

(b) $F \leq 0.3 \times 196 = 58.8$

Using 0.3×196

M1

If $P = 80$, $F = 58.8$

58.8 as answer

A1

If $P = 40$, $F = 40$

40 as answer

A1

3

(c) $P - 58.8 = 20 \times 0.8$

Three term equation of motion including 58.8

M1

Correct equation

A1

$P = 74.8$

A1

Correct P

A1

3

(d) $a = \frac{-58.8}{20} = -2.94$

Use of $F = ma$ with ± 58.8

M1

Correct acceleration with a negative sign

A1

$$0^2 = 6^2 + 2 \times (-2.94)s$$

Use of $v^2 = u^2 + 2as$ with $v = 0$

m1 A1

$$s = \frac{36}{5.88} = 6.12$$

Correct distance

A1

5

[12]

Q12.

(a) $0.5 = \frac{1}{2} \times a \times 4$

Use of $s = ut + \frac{1}{2}at^2$ with $u = 0$

M1

$$a = 0.25$$

Correct acceleration

A1

2

(b) $6 \times 9.8 - T = 6 \times 0.25$

*Three term equation of motion for particle,
with correct use of g*

M1

Correct equation

A1ft

$$T = 57.3$$

ag *Correct T from correct working*

A1ft

3

(c) $57.3 - F = 10 \times 0.25$

Three term equation of motion for the block

M1

Correct equation

A1ft

$$F = 54.8$$

Correct F

A1ft

$$R = 10 \times 9.8 = 98$$

R = 98 seen in working

B1

$$54.8 = 98\mu$$

Use of $F = \mu R$

m1

$$\mu = \frac{54.8}{98} = 0.559$$

correct μ

A1ft

6

[11]

Q13.

(a) $R = 60 \times 9.8 \times \cos 40^\circ = 450 \text{ N}$

Resolving perpendicular to the plane

EXAM PAPERS PRACTICE

M1

Correct reaction force

A1

2

(b) $F = 60 \times 9.8 \times \cos 50^\circ = 378$

Resolving parallel to the plane

M1

Correct F

A1

$$F \leq \mu R$$

Use of $F \leq \mu R$ or $F = \mu R$

m1

$$\mu \geq 0.839$$

Correct inequality

Accept 0.840

A1

4

(c) $F = 0.2 \times 60 \times 9.8 \cos 40^\circ$

$$60a = 60 \times 9.8 \cos 50^\circ - 0.2 \times 60 \times 9.8 \cos 40^\circ$$

Use of $F = \mu R$ to find F but not with $R = 60g$

M1

Three term equation of motion

M1

Correct equation

A1

$$a = 4.80 \text{ ms}^{-2}$$

Correct acceleration

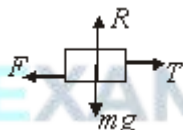
A1

5

[11]

Q14.

(a)



Correct diagram Papers Practice. All rights reserved.

B1

1

(b) $R = 4 \times 9.8 = 39.2 \text{ N}$

Correct normal reaction

B1

1

(c) $T - 4 \times 39.2 = 4 \times 2$

Three term equation of motion

M1

Correct equation

A1

$$T = 23.7 \text{ N (to 3sf)}$$

Correct T

A1

3

(d) $20 - 0.4 \times 39.2 = 4a$

M1

$$a = 1.08 \text{ ms}^{-2}$$

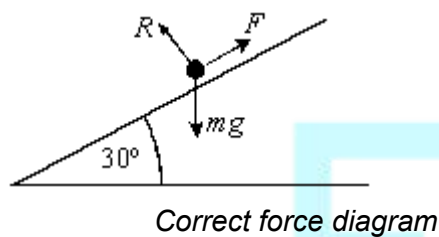
A1

2

[7]

Q15.

(a)



B1

1

(b) $R = 5 \times 9.8 \cos 30^\circ$

Finding normal reaction

M1

$$F = 0.2 + 5 \times 9.8 \cos 30^\circ = 8.5 \text{ N (to sf)}$$

Finding friction

M1

Correct friction

A1

3

(c) $5 \times 9.8 \sin 30^\circ - 0.2 \times 5 \times 9.8 \cos 30^\circ = 5a$

Three term equation of motion

M1

Correct equation

A1

$$a = \frac{5 \times 9.8 \sin 30^\circ - 0.2 \times 5 \times 9.8 \cos 30^\circ}{5}$$

Solving for a

M1

$$= 3.20 \text{ ms}^{-2}$$

Correct a

A1

4

(d) $v^2 = 0^2 + 2 \times 3.203 \times 1.2$

Constant acceleration equation with u = 0

M1

Correct equation

A1 ft.

$$v = 2.77 \text{ ms}^{-1}$$

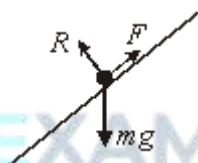
A1 ft.

3

[11]

Q16.

(a)



Correct force diagram

No extra forces

B1

1

(b) (i) $R = 80 \times 9.8 \cos 30^\circ = 679.0 \text{ N}$

Resolving perpendicular to the slope

Correct Reaction force

M1 A1

2

(ii) $F = 0.3R = 204 \text{ N}$

Multiply by 0.3

ag *Correct answer obtained*

M1 A1

2

- (c) $80a = 80 \times 9.8 \sin 30^\circ - 203.7$
Three term equation of motion
Correct equation

M1 A1

$a = 2.35 \text{ ms}^{-2}$
Solving for a
awrt 2.35

m1 A1

4

[9]

Q17.

- (a) $5^2 = 1^2 + 2 \times 4a$
Using constant acceleration equation to find a

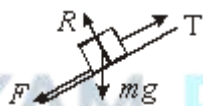
M1 A1

$a = 3 \text{ ms}^{-2}$
Correct equation

A1

3

- (b) (i)



Correct force diagram

B1

1

- (ii) $R = 20 \times 9.8 \cos 30^\circ = 170 \text{ N}$
Finding the normal reaction

M1

$F = 0.4 \times 20 \times 9.8 \cos 30^\circ = 67.9 \text{ N}$
Using $F = \mu R$

M1 A1

3

- (iii) $T - 67.9 - 20 \times 9.8 \sin 30^\circ = 20 \times 3$
Four term equation of motion

Correct equation

M1 A1

$$T = 226 \text{ N}$$

Solving for T

Correct T

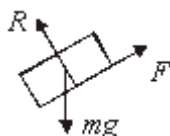
m1 A1

4

[11]

Q18.

(a)



Correct force diagram

B1

1

(b) $R = 7 \times 9.8 \cos 30^\circ = 59.4 \text{ N}$

Resolving perpendicular to plane

Correct force

M1/A1

2

(c) $F = 7 \times 9.8 \sin 30^\circ = 34.3 \text{ N}$

Resolving parallel to plane

Correct force

M1/A1

2

(d) $34.3 \leq \mu \times 59.4$

Using $F = \mu R$ or $F \leq \mu R$

M1

$$\mu \geq 0.577$$

Correct inequality

A1

2

(e) Remains at rest, as mass appears in both F and R , so inequality unchanged.

Conclusion

Reason

B1

B1

2

[9]

Q19.

(a) $T - 0.8 \times 5 \times 9.8 = 5a$

Equation of motion for one particle

M1

$$T = 5a + 39.2$$

Correct equation

A1

$$3 \times 9.8 - T = 3a$$

Equation of motion for other particle

M1

$$29.4 - (5a + 39.2) = 3a$$

Correct equation

A1

$$-9.8 = 8a$$

Solving for a

M1

$$a = -1.225 \text{ ms}^{-2}$$

Correct a

A1

6

(b) (ii) $T = 5a + 39.2$

Substituting and solving for T

M1

$$T = 33.1 \text{ N}$$

Correct T

A1

2

(c) $0^2 = 2^2 + 2 \times (-1.225)s$

*Use of CA equation with $v = 0$
Correct equation (allow $a = 1.225$)*

M1/A1

$$s = \frac{4}{2.45} = 1.63 \text{ m}$$

*Correct answer from correct working
(allow 1.64)*

A1

3

[11]



EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.